

CSCI2750: Assignment 2

Due: 11:59pm, February 20, 2026; Submit via Gradescope.

INSTRUCTIONS:

1. Answer all questions.
2. Please submit online through **Gradescope** and **assign pages to each question**.
3. By submitting this assignment, you are assumed to have read and understood the course assessment policy, which you can find on Blackboard. You are also assumed to have read and understood CUHK's guideline on academic honesty and on the use of AI tools.
4. All problems are expected to be solved by using the contents in this course. If you have to bring in results out of this scope, prior consent from the instructor shall be sought and a proper citation has to be provided. Exceptions are basic mathematical results in basic calculus and linear algebra.
5. You may discuss with your peers and tutors on these problems. But you must use your own wording and understanding to write up your solution. Similar expressions, equation, sentences, etc., may invite our investigation to turn your submission into an actionable disciplinary case.
6. You may present your work by scans of handwritten pages, or by any tablet-writing applications. However, please keep your submission legible and readable (otherwise we can't grade you...). Also, keep your submission in a single PDF file for the graders'/tutors' convenience.

Problem 1. (20%) Given a collection of n -dimensional vectors $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(m)}$, and a new instance $\mathbf{p} \in \mathbb{R}^n$; $\mathbf{x}^*$ is said to be the nearest neighbour of $\mathbf{p}$ if

$$d(\mathbf{x}^*, \mathbf{p}) \leq d(\mathbf{x}^{(i)}, \mathbf{p}) \quad \text{for all } i = 1, 2, \dots, m$$

when $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2$ is the Euclidean distance. Similarly, $\mathbf{x}^\dagger$ is said to be the *angle nearest neighbour* of $\mathbf{p}$ if

$$g(\mathbf{x}^\dagger, \mathbf{p}) \leq g(\mathbf{x}^{(i)}, \mathbf{p}) \quad \text{for all } i = 1, 2, \dots, m;$$

where

$$g(\mathbf{a}, \mathbf{b}) = \arccos \left(\frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2} \right).$$

is the angle between $\mathbf{a}$ and $\mathbf{b}$.

- (a) State an example dataset $\{\mathbf{x}^{(i)}\}$ and an instance $\mathbf{p}$ in 2D such that $\mathbf{x}^* \neq \mathbf{x}^\dagger$. Justify by a geometric illustration.
- (b) Suppose $\|\mathbf{p}\|_2 = \|\mathbf{x}^{(i)}\|_2 = 1$ for $i = 1, \dots, m$. Show that the nearest neighbour of $\mathbf{p}$ is always the *angle nearest neighbour* of $\mathbf{p}$.

Problem 2. (30%)

- (a) Show that the Manhattan distance

$$d_1(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1$$

satisfies the four common properties of dissimilarity measures (making it a metric), as listed on p.28 of Lecture 3.

- (b) In the lecture we've mentioned that convexity of a function
- f
- can be checked by the definition

$$f(\alpha \mathbf{x}_1 + (1 - \alpha) \mathbf{x}_2) \leq \alpha f(\mathbf{x}_1) + (1 - \alpha) f(\mathbf{x}_2).$$

Please show that the non-negative weighted sum of convex functions function $f_1, \dots, f_K$:

$$f(\mathbf{x}) = \sum_{k=1}^K c_k f_k(\mathbf{x}), \quad c_k \geq 0 \text{ for } k = 1, \dots, K,$$

is a convex function by definition.

- (c) Let
- $\mathbf{r} \in \mathbb{R}^n$
- ,
- $\mathbf{A} \in \mathbb{R}^{n \times n}$
- be pre-defined system parameters. Set
- $\mathbf{G} = \mathbf{A}^\top \mathbf{A}$
- . Show that

$$f(\mathbf{x}) = e^{\mathbf{x}^\top \mathbf{G} \mathbf{x} - 2\mathbf{r}^\top \mathbf{A} \mathbf{x} + \|\mathbf{r}\|_2^2}$$

is a convex by considering convex-preserving operations. That is, find the K -layers composition

$$f(\mathbf{x}) = f_1(f_2(\dots(f_K(\mathbf{x}))\dots))$$

in which all f_i 's are convex.

Problem 3. (20%) Consider the optimization problem:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & \frac{1}{4}x_1^4 + \frac{3}{2}x_2^2 \\ \text{s.t.} \quad & x_1^2 + x_2^2 \leq 1. \end{aligned} \tag{P}$$

- (a) Write down the optimal solution to (P) without solving it. (Remember to check for feasibility!)
- (b) Convert (P) into the standard form. Specifically, identify $f(\mathbf{x})$ and $g(\mathbf{x})$. Argue why are they convex. The derivative test is disallowed.
- (c) Write down the Lagrangian $\mathcal{L}(\mathbf{x}, \boldsymbol{\alpha})$ and the KKT conditions for (P). Then, find an optimal solution to (P) by solving the KKT conditions.
- (d) Write MATLAB + CVX code to check your answers in the above calculation.

Problem 4. (30%) This is a practical problem designed to sharpen your sense in data collection, and in data processing. You are required to prepare a toy dataset in the standard form. Specifically,

we will represent a corpus of m Wikipedia pages as an $m \times n$ matrix $\mathbf{X}$, where each row corresponds to a document and each column corresponds to a keyword in your dictionary:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}^{(1)\top} \\ \mathbf{x}^{(2)\top} \\ \vdots \\ \mathbf{x}^{(m)\top} \end{bmatrix} = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,n} \\ x_{2,1} & x_{2,2} & \dots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m,1} & x_{m,2} & \dots & x_{m,n} \end{bmatrix}$$

where $\mathbf{x}^{(i)} \in \mathbb{R}^n$ is the bag-of-words (BoW) vector of document i , and $x_{i,j}$ is the count of keyword j in document i . Follow the instructions below:

Step 1. Search for the Wikipedia pages of five modern scientists, and five famous athletes. All ten individuals should be notable in the last ~ 50 years. Record the exact page URLs and the date/time accessed for reproducibility.

Step 2. With the goal of making scientist vectors distinguishable from athlete vectors, *manually* choose seven keywords to form your dictionary. The keywords should not contain names (e.g., “Einstein”, “Messi”, “Ronaldo”) or obvious profession words (“scientist”, “footballer”). Try to use more content-bearing terms that you would expect to be more popular among the pages from the same group.

Step 3. Count the number of keyword occurrences in each Wikipage for each chosen keywords. Use a consistent procedure (e.g., by just counting the occurrence by “Ctrl+F”-ing the keywords) on each page. Then, compile the matrix $\mathbf{X} \in \mathbb{R}^{10 \times 7}$ with the raw counts $x_{i,j}$.

Step 4. On MATLAB, compile the (pair-wise) dissimilarity matrix

$$\mathbf{D} = \begin{bmatrix} d(\mathbf{x}^{(1)}, \mathbf{x}^{(1)}) & d(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) & d(\mathbf{x}^{(1)}, \mathbf{x}^{(3)}) & \dots & d(\mathbf{x}^{(1)}, \mathbf{x}^{(10)}) \\ d(\mathbf{x}^{(2)}, \mathbf{x}^{(1)}) & d(\mathbf{x}^{(2)}, \mathbf{x}^{(2)}) & d(\mathbf{x}^{(2)}, \mathbf{x}^{(3)}) & \dots & d(\mathbf{x}^{(2)}, \mathbf{x}^{(10)}) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d(\mathbf{x}^{(10)}, \mathbf{x}^{(1)}) & d(\mathbf{x}^{(10)}, \mathbf{x}^{(2)}) & d(\mathbf{x}^{(10)}, \mathbf{x}^{(3)}) & \dots & d(\mathbf{x}^{(10)}, \mathbf{x}^{(10)}) \end{bmatrix},$$

where

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2$$

and generate a corresponding heatmap of $\mathbf{D}$ with a colorbar, clear title, and labeled axes (Wikipage name). You are disallowed from using external functions like `pdist()`. The implementation should be *from scratch*.

Step 5. Write a brief report (≤ 4 pages) to summaries your findings. Cover:

- i) A list of the 10 pages (with URLs and access date/time)
- ii) Your chosen dictionary (the 7 keywords) with a rationale for choosing each term
- iii) The collected matrix $\mathbf{X}$ (in table form or in screenshot)
- iv) The compiled dissimilarity matrix $\mathbf{D}$ and its heatmap
- v) A brief take-away message on what you have learnt in attempting this problem
- vi) The MATLAB codes (with proper comment) used in your report as an appendix

— THE END —